

Passive Acoustic Monitoring (PAM)

This page collects field-tested practices for **flying a Slocum quietly** and for the **mission design** PAM work needs. It is platform-focused on Slocum; some underlying principles (reduce pump activity, silence the altimeter) apply to any glider.

Source

Paraphrased from the UG2 community Slack (which has a dedicated PAM channel) and the Teledyne Webb Research user forum. Reflects field experience from NOAA Fisheries, UVI, AMLR, Rutgers, WHOI, and others. This is a condensed community reference — it is **not** a manufacturer document. For sensor integration, work directly with Teledyne and your instrument vendor.

Flying quietly: the self-noise budget

The biggest wins come from reducing the glider's own noise. In order of impact:

1. Buoyancy pump — the loudest thing on the glider

Every inflection runs the pump, which dominates the spectrogram (the piston shows as bright broadband bars). To reduce it:

- **Minimize pump activity:** fewer, deeper, slower inflections; use the minimum buoyancy drive that still flies. The same levers as the [Power Saving](#) page help here — quieter usually means lower power too.
- **Drift segments:** where the science allows, drift-at-depth (wide deadband) keeps the pump off for long quiet stretches.

2. Pitch control — prefer "set once" over constant servo

Constantly servoing the pitch battery makes noise. A common PAM approach:

- First let the glider **find trim** with **autoballast on servo** (optimal pitch and oil/ballast settings).
- Then switch to `battpos` / "**set once**" so the battery moves **once per half-yo** to reach the target and then holds — much quieter than continuous adjustment.